

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

The equation $x^2 + (y - 2)^2 = 36$ represents circle A. Circle B is obtained by shifting circle A down 4 units in the xy-plane. Which of the following equations represents circle B?

Answer

- A. $x^2 + (y + 2)^2 = 36$
- B. $x^2 + (y - 6)^2 = 36$
- C. $(x - 4)^2 + (y - 2)^2 = 36$
- D. $(x + 4)^2 + (y - 2)^2 = 36$

Correct Answer: A

Rationale

Choice A is correct. The standard form of an equation of a circle in the xy-plane is $(x - h)^2 + (y - k)^2 = r^2$, where the coordinates of the center of the circle are (h, k) and the length of the radius of the circle is r . The equation of circle A, $x^2 + (y - 2)^2 = 36$, can be rewritten as $(x - 0)^2 + (y - 2)^2 = 6^2$. Therefore, the center of circle A is at $(0, 2)$ and the length of the radius of circle A is 6. If circle A is shifted down 4 units, the y-coordinate of its center will decrease by 4; the radius of the circle and the x-coordinate of its center will not change. Therefore, the center of circle B is at $(0, 2 - 4)$, or $(0, -2)$, and its radius is 6. Substituting 0 for h , -2 for k , and 6 for r in the equation $(x - h)^2 + (y - k)^2 = r^2$ yields $(x - 0)^2 + (y - (-2))^2 = (6)^2$, or $x^2 + (y + 2)^2 = 36$. Therefore, the equation $x^2 + (y + 2)^2 = 36$ represents circle B.

Choice B is incorrect. This equation represents a circle obtained by shifting circle A up, rather than down, 4 units.

Choice C is incorrect. This equation represents a circle obtained by shifting circle A right, rather than down, 4 units.

Choice D is incorrect. This equation represents a circle obtained by shifting circle A left, rather than down, 4 units.